Project Proposal

Bright Path

Learning Management System

GROUP : WD_55_2.2
MODULE : IT3040

Introduction

BrightPath is a comprehensive, web-based Learning Management System (LMS) specifically engineered for the university student community. It serves as a centralized hub designed to bridge the gap between academic resource availability and student accessibility through a structured, peer-to-peer knowledge-sharing model.

Aims & Objectives

Aim

- To develop a centralized, secure, and smart Peer-to-Peer Learning Management System that optimizes academic resource sharing and collaborative learning among university students.

Objective

- **Requirement Analysis:** Identify student needs for academic resources and live sessions.
- **System Architecture:** Design a scalable platform using Django and Vue.js.
- **Secure Access**: Implement JWT-based authentication for verified student login.
- **Content Hub:** Build APIs for organized video and past paper management.
- **Live Sync:** Integrate Google Classroom API for synchronized "Kuppi" sessions.
- **Gamification:** Develop a Point-Based Reward system for high-quality sharing.
- **Smart Alerts:** Utilize Celery & Redis for automated Gmail notifications.
- **Testing & Deployment:** Ensure system reliability through rigorous unit and integration testing.

Practical Feasibility & Values

- ❑ **Modern Tech Stack**: Using reliable Django and Vue.js frameworks.
- ❑ **Cloud Storage:** Scalable data management using AWS S3.
- ❑ **Open-Source**: Cost-effective development with no licensing fees.
- ❑ **API Integration**: Easy connection with Google Classroom services

Value to Students:

- ✓ Time Saving
- ✓ Quality Content
- ✓ Peer Learning
- ✓ Personalized

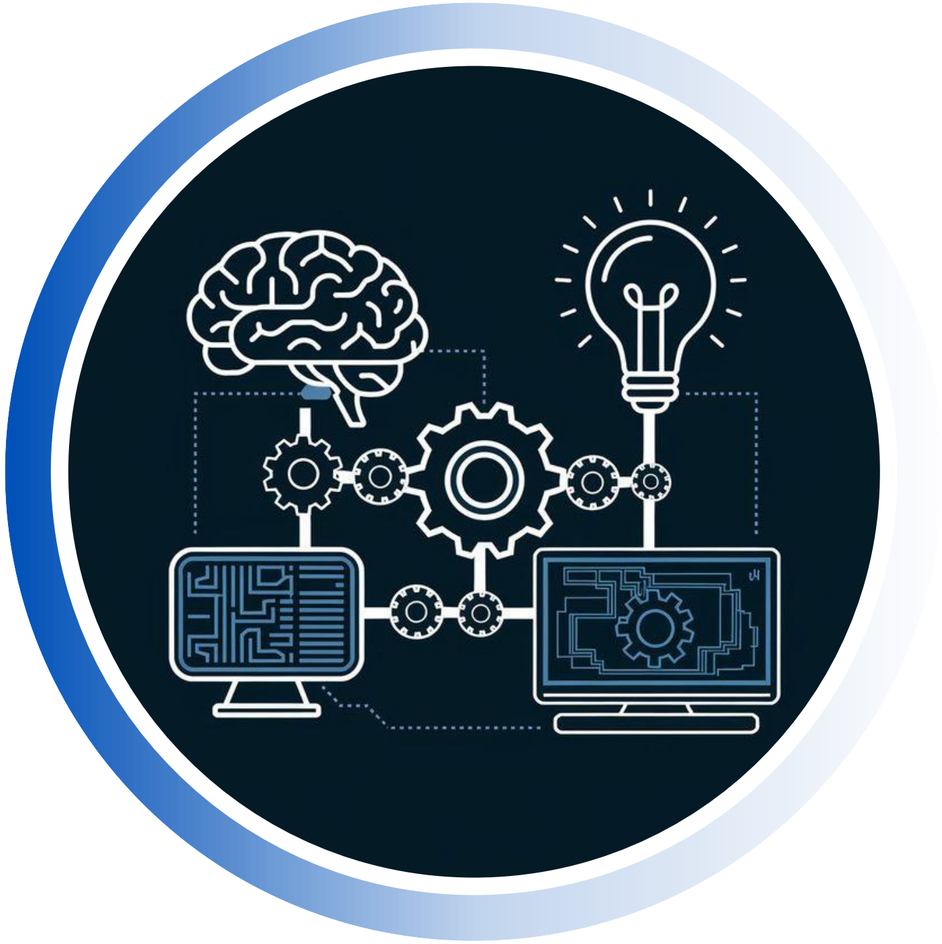
Target Users

- University Students
- Academic Staff / Teachers
- System Administrators

Tools and Technologies

- Vue.js
- Django (Python)
- Google Classroom API
- PostgreSQL
- Redis & Celery

Project Proposal



Functions

Identity & Access Management

- **Custom User Model :** Managing distinct access levels for Students, Staff/Faculty, and Administrative roles.
- **Security:** Implemented SimpleJWT for secure, stateless authentication between Vue.js and Django.
- **Verification Logic:** Backend service to validate Student identity against university criteria.
- **Dynamic Dashboard:** Tailored API to filter academic content based on the user's specific Year and Stream.

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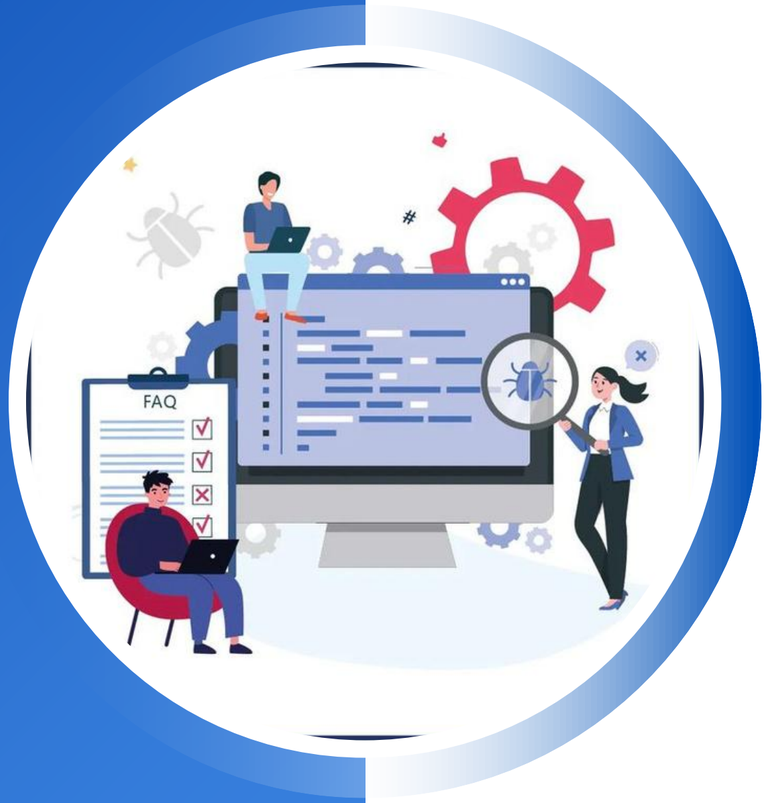
Content & Live Integration



- **Resource Management:** Developed secure CRUD endpoints for managing Lecture Videos and Past Papers.
- **Cloud Hosting:** Integrated AWS S3 with django-storages for high-performance media storage.
- **Google API Sync:** Integrated Google Classroom for automated "Kuppi" session scheduling and synchronization.
- **Structured Delivery:** Modular API design to categorize content into Videos, Notes, and Papers.

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Library & Gamification

- **Library Engine:** Built a centralized PDF repository for high-speed resource sharing.
- **Ranking Logic:** Automated quality control system based on user ratings and relevance weights.
- **Point System:** Developed a Gamified Wallet using Django Signals to reward quality contributors.
- **Search Optimization:** Utilized Django Q objects for advanced filtering and high-speed search results.

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Automation & Intelligence

- **Asynchronous Tasks:** Implemented Celery & Redis for efficient background process management.
- **Real-time Alerts:** Automated Gmail notification system for real-time updates on new resource uploads.
- **Automated Triggers:** Scheduled "Delayed Tasks" for automated user feedback and review requests.
- **Recommendation Engine:** Smart algorithm to suggest modules based on peer activity within the same stream.

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Conclusion

BrightPath successfully provides a centralized, secure, and smart ecosystem designed to revolutionize the university learning experience. By integrating a robust **Django** and **Vue.js** architecture, the platform effectively bridges the gap between resource availability and student accessibility through its automated ranking system and verified identity management. Beyond being a mere repository, BrightPath fosters a collaborative culture by incentivizing high-quality peer-to-peer sharing and synchronizing live "Kuppi" sessions, ultimately offering a scalable and efficient solution for modern academic challenges.

Any Questions?

BrightPath

Thank You

For your attention to this presentation.